Week 7 — Case Study: Estimating Basin Depth in Nevada using FFTs

Geophysics 3A: Potential Fields & Geothermics

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1 Purpose of the Demonstration

This demonstration extends the basic ideas of Fourier transforms developed in Demo 1 to spatial potential-field data. The focus is on understanding how sampling, filtering, and physical assumptions control what geological information can be recovered from gravity data.

The notebook progresses deliberately from:

- synthetic geological models with known truth,
- to real Complete Bouguer gravity data from Nevada,
- to physically motivated removal of regional signals via isostatic correction,
- and finally to spectral interpretation and inversion.

2 In-Class Demonstration Outline

A suggested walkthrough is:

1. **Nevada Bouguer gravity:** identify basin-scale stripes.
2. **Isostatic correction:** remove long-wavelength crustal signal.
3. **Spectral peaks:** identify dominant basin wavelengths.
4. **Aliasing pipeline:** demonstrate failure of inversion under coarse sampling.

3 Spatial Fourier Transform

For a uniformly sampled profile, the discrete Fourier transform gives

$$\hat{g}(k_n) = \sum_{j=0}^{N-1} g(x_j)\, e^{-i2\pi k_n x_j}, \tag{1}$$

with discrete wavenumbers

$$k_n = \frac{n}{N\Delta x}. \tag{2}$$

The amplitude spectrum $|\hat{g}(k)|$ shows how much power exists at each wavelength. For potential fields, spectra typically show a *red-noise* background: power increases systematically toward longer wavelengths.

4 Aliasing in the Spectral Domain

When the sampling interval increases:

- the Nyquist wavelength moves to longer scales,

- true basin-scale power migrates to incorrect longer wavelengths,
- spurious spectral peaks appear that look statistically significant.

The Fourier transform faithfully represents the sampled data, even when that data are aliased.

5 Airy Isostatic Gravity Correction

The long-wavelength Bouguer gravity field over Nevada is dominated by isostatic compensation of crustal thickness variations. Under Airy isostasy, surface loads are compensated by a crustal root at depth T_0.

In the spectral domain, the gravity effect of a density interface at depth T_0 is

$$\hat{g}_{\mathrm{root}}(\mathbf{k}) = 2\pi G\,\Delta\rho\,\hat{h}(\mathbf{k})\,e^{-|\mathbf{k}|T_0}, \tag{3}$$

where:

- $\Delta\rho$ is the density contrast across the Moho,
- $e^{-|\mathbf{k}|T_0}$ is the upward-continuation filter.

Subtracting this predicted root contribution from the observed Bouguer gravity yields the *isostatic residual*:

$$g_{\mathrm{iso}} = g_{\mathrm{Bouguer}} - g_{\mathrm{root}}. \tag{4}$$

This is a physically motivated, band-limited removal of long-wavelength power, not a statistical detrend.

6 Spectral Interpretation of Nevada Gravity

After isostatic correction:

- the mean gravity is near zero,
- long-wavelength crustal signals are suppressed,
- basin-scale anomalies (20–60 km) are enhanced.

Mean E–W and N–S power spectra reveal:

- elevated E–W power at 20–40 km wavelengths,
- longer characteristic wavelengths in the N–S direction,
- a red-noise background consistent with potential-field theory.

7 Statistical Significance of Spectral Peaks

Because adjacent profiles are spatially correlated, the number of independent spectral estimates is smaller than the number of rows or columns. An effective number of degrees of freedom is estimated from the spatial autocorrelation length.

After fitting and removing a power-law background, a chi-squared test is used to identify statistically significant spectral peaks. For Nevada:

- the dominant E–W peak lies near $\lambda \approx 20$–25 km,
- consistent with Basin-and-Range basin spacing.

8 High-Pass Filtering and Inversion

Before inverting gravity for sediment thickness, residual long-wavelength power must be removed. A raised-cosine high-pass filter is applied:

- wavelengths > 100 km are blocked,
- wavelengths < 60 km are passed,
- a smooth taper prevents Gibbs ringing.

The filtered gravity is inverted using the Parker–Oldenburg method, which relates gravity to interface relief in the spectral domain.

9 Impact of Aliasing on Inversion

Repeating the full processing pipeline on decimated data shows that:

- when the Nyquist wavelength exceeds basin width,
- sediment thickness maps degrade rapidly,
- artefacts appear that have no geological meaning.

No inversion algorithm can recover information lost to aliasing.

10 Sediment Thickness Inversion and Spectral Filtering

After removal of the long-wavelength isostatic signal, the remaining gravity field primarily reflects density variations in the shallow crust. In the Basin-and-Range Province, these variations are dominated by low-density sediment filling extensional basins over denser crystalline basement. We now describe how this signal is converted into a sediment thickness model, and why additional spectral filtering is required before inversion.

10.1 Physical Meaning of the Sediment Thickness Model

The sediment thickness model $h(x, y)$ represents the vertical thickness of low-density basin fill that replaces higher-density basement rock. The relevant density contrast is

$$\Delta\rho = \rho_{\text{basement}} - \rho_{\text{sediment}}, \tag{5}$$

which is taken here to be approximately 670 kg m^{-3}.

Under this convention:

- negative gravity anomalies correspond to thick sedimentary basins,
- positive anomalies correspond to basement highs or exposed ranges.

The model treats sediment as a single layer with uniform density overlying basement. This is a deliberate simplification. Gravity is most sensitive to the largest density contrast, and cannot resolve fine-scale stratigraphy within the sediment column. The resulting sediment thickness therefore captures the dominant mass deficit associated with basin fill, rather than detailed internal layering.

10.2 Forward Gravity Model for an Interface

For an interface with mean depth z_0, relief $h(x, y)$, and density contrast $\Delta\rho$, the linearized gravity response in the Fourier domain is

$$\hat{g}(\mathbf{k}) = 2\pi G \Delta\rho \, e^{-|\mathbf{k}|z_0} \, \hat{h}(\mathbf{k}), \tag{6}$$

where $\mathbf{k}$ is the horizontal wavenumber vector and $e^{-|\mathbf{k}|z_0}$ is the upward-continuation kernel.

This expression highlights two fundamental properties of gravity data:

- short wavelengths decay rapidly with depth,
- gravity observations are inherently band-limited by source depth.

10.3 Instability of Direct Inversion

Formally inverting the linear relation gives

$$\hat{h}(\mathbf{k}) = \frac{\hat{g}(\mathbf{k})}{2\pi G \Delta\rho}\, e^{|\mathbf{k}|z_0}. \tag{7}$$

The exponential factor $e^{|\mathbf{k}|z_0}$ amplifies high-wavenumber components. As a result:

- noise,
- discretization error,
- aliasing from inadequate sampling,

are dramatically magnified. This makes naive downward continuation unstable and physically meaningless.

10.4 The Parker–Oldenburg Method

The Parker–Oldenburg method (Parker, 1973; Oldenburg, 1974) addresses this instability by treating the gravity–interface relationship as weakly nonlinear and solving iteratively in the spectral domain.

In practice, the method:

1. starts with an initial guess for the interface relief $h(x, y)$,
2. forward-calculates the gravity anomaly,
3. compares predicted and observed gravity,
4. updates the interface estimate,
5. iterates until convergence.

Each iteration uses fast Fourier transforms, with computational cost $O(N \log N)$, making the method efficient even for large grids. The result is a stable estimate of interface relief, provided the input gravity is appropriately filtered.

10.5 Why High-Pass Filtering Is Required

Even after isostatic correction, the gravity field may still contain long-wavelength signals arising from:

- imperfect isostatic compensation,
- lithospheric flexure,
- deeper crustal or mantle density variations.

If these wavelengths are left in the data, the inversion will interpret them as extremely thick sediment layers, producing unphysical basin depths.

High-pass filtering therefore encodes a geological prior:

> sediment thickness varies on wavelengths comparable to basin widths, not on lithospheric scales.

10.6 Raised-Cosine High-Pass Filter

A sharp spectral cutoff produces Gibbs ringing in the spatial domain, manifesting as oscillations and spurious basin edges. To avoid this, a smooth raised-cosine (Hann) filter is applied:

$$H(k) = \begin{cases} 0, & k \le k_{\mathrm{lo}}, \\ \frac{1}{2}\left[1 - \cos\left(\pi \frac{k - k_{\mathrm{lo}}}{k_{\mathrm{hi}} - k_{\mathrm{lo}}}\right)\right], & k_{\mathrm{lo}} < k < k_{\mathrm{hi}}, \\ 1, & k \ge k_{\mathrm{hi}}. \end{cases} \tag{8}$$

In this application:

- wavelengths > 100 km are fully suppressed,
- wavelengths < 60 km are fully passed,
- intermediate wavelengths are smoothly tapered.

This filter removes residual regional power while preserving basin-scale structure, and ensures that the Parker–Oldenburg inversion remains stable and geologically plausible.

10.7 Interpretive Context

It is important to emphasize that:

- the inversion is not unique,
- the sediment thickness model is conditional on sampling, filtering, and density assumptions,
- no inversion can recover wavelengths shorter than the Nyquist limit.

The success of the Parker–Oldenburg inversion in this case reflects the fact that sampling, isostatic correction, and spectral filtering have already constrained the problem to a physically interpretable band of wavelengths.

10.8 Key Takeaway

The Parker–Oldenburg method is best understood as a physically motivated, regularized downward continuation. It produces meaningful sediment thickness estimates only when the gravity data have been appropriately sampled, filtered, and interpreted.